

TP3 : DOCKER

Dans ce TP, l'objectif était de découvrir les bases de Docker, la gestion des conteneurs, des volumes, des réseaux ainsi que l'exécution d'une base de données MySQL dans un environnement totalement isolé. L'ensemble des manipulations a permis de comprendre les principes fondamentaux de la conteneurisation ainsi que les avantages de l'isolation, de la persistance et du déploiement simplifié.

On a tout d'abord commencé par installer docker sur la machine.
on a installer curl on a creer un fichier text appelle hello.txt

on a fait docker run -it alpine sh

```
echo "hello world" > helloworld.txt
docker run -it alpine sh
docker run -it -v $(pwd):/app alpine sh
ls /app
on a tous les fichiers de mon rep qui s'affiche
```

pour creer un volume on fait :
docker volume create mon volume
pour lister :
docker volume ls
docker volume rm monvolume (pour supprimer)

ensuite :
docker volume create monv
docker run -it -v monv:/data alpine sh
ls
on remarqu'un repertoire data a ete creer
on creer :
/ # echo "Bonjour ecrit a l'interieur du conteneur" >/data/hello.txt

```
/data # exit
ing@ing:~$ docker ps -a
```

```
docker rm ab04e2a83943
```

Ici, on demande a docker de lancer un serveur Mysql version 8 dans un conteneur. Donc e conteneur sera un MySql complet, identique à celui qu'on installe sur une machine normale.

On passe ensuite, on passe une variable d'environnement au serveur Mysql, on ne peut pas demarrer sans ce mdp.

—name mysql donne un nom au conteneur

-v mysql_data:/var/lib/mysql : on crée un volume persistant. On dit de ne pas stocker les données de mysql dans le conteneur mais de les stocker dans un volume docker appelé mysql_data.

Comme ça quand on supprime le conteneur les données sont toujours présentes et comme ça si on veut supprimer un conteneur et en créer un autre les données seront toujours là.

```
docker run -d --name mysql \
-e MYSQL_ROOT_PASSWORD=pass \
-v mysql_data:/var/lib/mysql \
mysql:8
```

```
ing@ing:~/2decembre$ mysql -u
mysql: [ERROR] mysql: option '-u' requires an argument.
ing@ing:~/2decembre$ docker exec -it mysql mysql -uroot -ppass
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.7 MySQL Community Server - GPL
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE demo;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> USE demo
Database changed
mysql> CREATE TABLE users (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO users (name) VALUES ('Alice'), ('Bob'), ('Charlie');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM users;
```

```
+----+-----+
| id | name  |
+----+-----+
| 1  | Alice |
```

```
| 2 | Bob   |
| 3 | Charlie |
+----+-----+
3 rows in set (0.00 sec)
```

mysql> EXIT

Bye

ing@ing:~/2decembre\$ docker stop mysql

mysql

ing@ing:~/2decembre\$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

ing@ing:~/2decembre\$ docker start mysql

mysql

ing@ing:~/2decembre\$ docker exec -it mysql mysql -uroot -ppass

mysql: [Warning] Using a password on the command line interface can be insecure.

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 8

Server version: 8.4.7 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE demo

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

mysql> SELECT * FROM users;

```
+----+-----+
| id | name   |
+----+-----+
| 1 | Alice  |
| 2 | Bob    |
| 3 | Charlie|
+----+-----+
3 rows in set (0.00 sec)
```

mysql> exit

Bye

ing@ing:~/2decembre\$ docker stop mysql

mysql

ing@ing:~/2decembre\$ docker run -d --name mysql2 \
-e MYSQL_ROOT_PASSWORD=pass \

```
-v mysql_data:/var/lib/mysql \
mysql:8
873204d45a5a87d955645a6ecea2a42260d762ee29edc59c5b735dffd727c8
ing@ing:~/2decembre$ docker exec -it mysql2 mysql -uroot -ppass
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.7 MySQL Community Server - GPL
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> SHOW DATABASES;
```

```
+-----+
| Database          |
+-----+
| demo              |
| information_schema|
| mysql             |
| performance_schema|
| sys               |
+-----+
5 rows in set (0.00 sec)
```

```
mysql> USE demo;
```

Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> SELECT * FROM users;
```

```
+----+-----+
| id | name  |
+----+-----+
| 1  | Alice |
| 2  | Bob   |
| 3  | Charlie|
+----+-----+
3 rows in set (0.00 sec)
```

```
mysql>
```

```
root@ing:~# cd 2decembre
```

```
-bash: cd: 2decembre: Aucun fichier ou dossier de ce type
root@ing:~# docker start mysql2
mysql2
root@ing:~# docker exec -it mysql2 mysql -uroot -ppass
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.4.7 MySQL Community Server - GPL
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> USE demon
ERROR 1049 (42000): Unknown database 'demon'
mysql> USE demo
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
mysql> SELECT * from users
-> ^C
mysql> SELECT * from users;
+----+-----+
| id | name  |
+----+-----+
| 1  | Alice |
| 2  | Bob   |
| 3  | Charlie |
+----+-----+
3 rows in set (0.01 sec)
```

```
mysql> exit
Bye
root@ing:~# docker stop mysql2
mysql2
root@ing:~# docker start mysql
mysql
root@ing:~# docker exec -it mysql mysql -uroot -ppass
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.4.7 MySQL Community Server - GPL
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> USE demo
```

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> SELECT * FROM users;
```

```
+----+-----+
```

```
| id | name |
```

```
+----+-----+
```

```
| 1 | Alice |
```

```
| 2 | Bob   |
```

```
| 3 | Charlie |
```

```
+----+-----+
```

3 rows in set (0.01 sec)

```
mysql> INSERT INTO users (name) VALUES ('Dora');
```

Query OK, 1 row affected (0.00 sec)

```
mysql> exit
```

Bye

```
root@ing:~# docker stop mysql
```

```
mysql
```

```
root@ing:~# docker start mysql2
```

```
mysql2
```

```
root@ing:~# docker exec -it mysql2 mysql -uroot -ppass
```

mysql: [Warning] Using a password on the command line interface can be insecure.

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 8

Server version: 8.4.7 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> USE demo
```

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

```
mysql> SELECT * from users;
```

```
+----+-----+
```

```
| id | name |
```

```
+----+-----+
```

```
| 1 | Alice |
```

```
| 2 | Bob   |
```

```
| 3 | Charlie |
```

```
| 4 | Dora   |
```

```
+----+-----+
```

4 rows in set (0.00 sec)

```
mysql> SHOW DATABASES;
```

```
+-----+
```

```
| Database |
```

```
+-----+
```

```
| demo      |
```

```
| information_schema |
```

```
| mysql      |
```

```
| performance_schema |
```

```
| sys        |
```

```
+-----+
```

5 rows in set (0.00 sec)

docker network ls

docker run -d --name web nginx

```
root@ing:~# docker run -d -p8080:80 nginx
```

```
8632a2e03e91876ce9e29192c40f8e11c9d529a4b96ce743c2e41375a8c8a1ac
```

```
root@ing:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
--------------	-------	---------	---------	--------	-------

NAMES

8632a2e03e91	nginx	"/docker-entrypoint...."	21 minutes ago	Up 21 minutes	
--------------	-------	--------------------------	----------------	---------------	--

0.0.0.0:8080->80/tcp, :::8080->80/tcp	musing_khorana				
---------------------------------------	----------------	--	--	--	--

2d28b6c8ebdc	nginx	"/docker-entrypoint...."	31 minutes ago	Up 31 minutes	80/tcp
--------------	-------	--------------------------	----------------	---------------	--------

site2

79ec784c74b0	nginx	"/docker-entrypoint...."	43 minutes ago	Up 43 minutes	80/tcp
--------------	-------	--------------------------	----------------	---------------	--------

web

873204d45a5a	mysql:8	"docker-entrypoint.s..."	4 hours ago	Up 3 hours	3306/tcp,
--------------	---------	--------------------------	-------------	------------	-----------

33060/tcp	mysql2				
-----------	--------	--	--	--	--

```
root@ing:~# docker exec -it 8632a2e03e91
```

"docker exec" requires at least 2 arguments.

See 'docker exec --help'.

Usage: docker exec [OPTIONS] CONTAINER COMMAND [ARG...]

Run a command in a running container

```
root@ing:~# nano Dockerfile
```

```
root@ing:~# nano Dockerfile
```

```
root@ing:~# docker build -t Dockerfile
```

invalid argument "Dockerfile" for "-t, --tag" flag: invalid reference format: repository name must be lowercase

See 'docker build --help'.

```
root@ing:~# docker build -tDockerfile
```

invalid argument "Dockerfile" for "-t, --tag" flag: invalid reference format: repository name must be lowercase

See 'docker build --help'.

```
root@ing:~# docker build -t Dockerfile
```

invalid argument "Dockerfile" for "-t, --tag" flag: invalid reference format: repository name must be lowercase

See 'docker build --help'.

```
root@ing:~# nano Dockerfile
```

```
root@ing:~# docker build -t mon-app .
```

ERRO[0000] Can't add file /root/.gnupg/S.dirmngr to tar: archive/tar: sockets not supported

ERRO[0000] Can't add file /root/.gnupg/S.gpg-agent to tar: archive/tar: sockets not supported

ERRO[0000] Can't add file /root/.gnupg/S.gpg-agent.browser to tar: archive/tar: sockets not supported

ERRO[0000] Can't add file /root/.gnupg/S.gpg-agent.extra to tar: archive/tar: sockets not supported

ERRO[0000] Can't add file /root/.gnupg/S.gpg-agent.ssh to tar: archive/tar: sockets not supported

Sending build context to Docker daemon 227.2MB

Step 1/2 : FROM alpine:latest

---> 706db57fb206

Step 2/2 : CMD echo "Bonjour Polytech Nancy"

---> Running in ede085094010

Removing intermediate container ede085094010

---> 8898cb1419a6

Successfully built 8898cb1419a6

Successfully tagged mon-app:latest

```
root@ing:~# docker run mon-app
```

Bonjour Polytech Nancy